

The goal of the project is to design an amplifier with the specifications in the table below:

Table 1: Design Specifications

Course	EE 140	EE 240A
Technology	Gpdk045	
Power Supplies	GND (0V), VDDL ($\leq 1.1V$), VDDH ($\leq 1.8V$)	
Closed Loop DC Gain	2	
Load	500 Ohms, 50 pF (output at capacitor)	
Settling Time (T_{settle}) $= \max\{T_{\text{settle, rise}}, T_{\text{settle, fall}}\}^1$	to be calculated for 1.4V output step rising and falling	
Total Error	$\leq 0.2\%$	
Power Consumption	$\leq 2.5 \text{ mW}$	$\leq 1.5 \text{ mW}$
Output Voltage Swing	$\geq 1.4V$	
Maximum Current Mirror Ratio	10	
Maximum Total Capacitance ²	10 pF	5 pF
Maximum Total Resistance	200 kOhm	100 kOhm
CMRR at DC	$\geq 70\text{dB}$	
PSRR at DC	$\geq 50\text{dB}$	
Phase Margin	$\geq 45^\circ$	
Figure of Merit (FoM) to Maximize	$\frac{1}{T_{\text{settle}} (\text{us}) \times P_{\text{total}} (\text{mW})}$	

To achieve the maximum settling time, according to the following statement, the maximum time should be less than 180 nanoseconds in order to be able to initialize all the pixels within one cycle.

amplifier to set an accurate pixel voltage within a short period of time. LCD displays commonly operate at a refresh rate of 60 Hz. In this project, you will be designing a driver for a 38-mm smartwatch display, which has 272 x 340 pixels. Your driver must drive *all pixels sequentially* in the display in one period of the refresh rate.

$$\max(T_{\text{rise}}, T_{\text{fall}}) = \frac{1}{\frac{60}{272 * 340}} \approx 180\text{ns}$$

After obtaining the parameters, the amplifier must be optimized so that the figure of merit (FOM), which is the inverse of power consumption multiplied by settling time, is maximized.

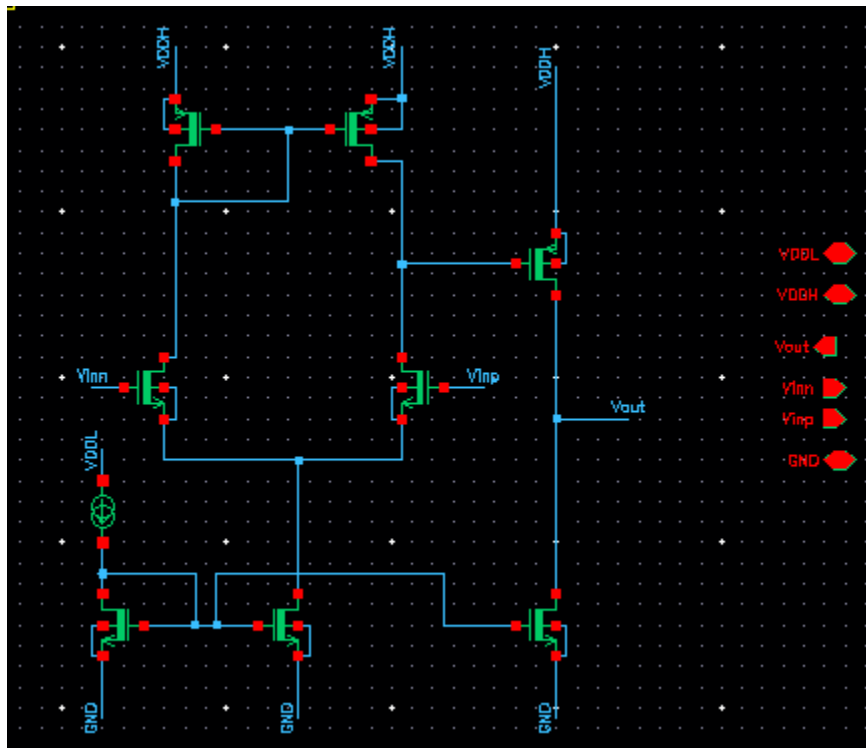
The power consumption should not exceed 1.5 mW, and since power is the product of current and voltage, and the voltage of the amplifier we are using is 1.8V (although we also use a 1.1V power supply, the current through it will be lower, which we need to consider), the maximum allowed current for the amplifier should not exceed 800 μA . The smaller the current, the larger the FOM will be, making the design more efficient. However, this can only be reduced up to the point where other parameters, such as the bandwidth threshold fufu, do not fall below 100 MHz, and the settling time remains under 180 nanoseconds.

The formulas and relationships are the same as those written in the file "PoleZero.d_Formula.docx."

We begin the design with the MATLAB code "DesignCascodeAmp.m" for maximum gain. The smaller V_{xVx} is, the higher the gain of the first stage will be, so we choose the smallest possible value for it. According to the MATLAB LUT matrices, the minimum value for V_{DSVDS} of the transistor is 0.05V, which is just enough to turn on the transistor, so we select this value. Thus, $V_{xVx}=50mV$.

The value of V_{cmVcm} , which is the DC input, should also be greater than V_{gsVgs} of the input transistor by at least the amount of V_{xVx} , and this is one of the output parameters of the MATLAB code. It might also require tuning to ensure that V_{xVx} remains at 50 mV.

The amplifier schematic will be as shown below:



Note: In order for the amplifier's phase to start at 180 degrees at frequency 0 or DC, I swapped the positions of V_{inn} and V_{inp} , as shown in the diagram above.

Regarding the choice of topology:

The Cascode topology was selected because, compared to the telescopic topology, it has the following advantages:

1. Fewer design parameters: It does not require separate bias voltages, making the design simpler.

2. Higher V_{DS} : Since the cascode topology uses two transistors in series (whereas the telescopic design uses four), a higher voltage can be applied across each transistor, making it easier for them to stay in the saturation region.
3. Pre-existing MATLAB code: We had already written MATLAB code for the analysis of the cascode amplifier, although it requires some minor adjustments.

MATLAB Code Explanation:

The code consists of three main parts:

1. Design of the first stage of the amplifier, which is similar to Exercise 5b.
2. Design of the second stage of the amplifier, which is similar to Exercise 5a.
3. Symbolic analysis of the two-stage amplifier, where values obtained from the previous two stages for g_m and r_{oro} are substituted, and calculations for phase margin, rise time T_{rise} , etc., are performed.

Regarding the second stage, it is important to note that the positions of the NMOS and PMOS transistors are reversed compared to Exercise 5a, and in the use of VLOOKUP commands, the N-channel (nch) and P-channel (pch) have been swapped.

Overall, this MATLAB code, for an input frequency of $f_u=100$ MHz and given values for g_m of both stages, transistor dimensions, and ultimately the Bode diagram, shows the performance of the amplifier.

Assuming $g_{m1}=g_{m2}=1 \times 10^{-4}$, the output of the program is as follows, which, as we can see, is not yet our desired result:

```

For gm1=100 (uA/v)

Vx=0.05 (v); Gain_1st_Stage=44.67; Opt_Vout1=1.00 (v); Vicm=0.48 (v); I1=5.88 (uA)

Wn1=3.17 (uM); Ln1=0.50 (uM); Wp1=0.23 (uM); Lp1=1.00 (uM);

For gm2=100 (uA/v)

Gain_2nd_Stage=49.06; Opt_Vout=0.95 (v); I2=5.88 (uA)

Wn2=0.09 (uM); Ln2=0.50 (uM); Wp2=2.88 (uM); Lp2=1.00 (uM);

Gain margin= 26.82 (dB) at 125.45 (MHz); Phase Margin= 64.47° at 7.1 (MHz)

      RiseTime: 6.6537e-08
TransientTime: 1.9952e-07
SettlingTime: 1.9954e-07
SettlingMin: -2.0342
SettlingMax: -1.8123|
Overshoot: 1.7569
Undershoot: 0.0516
      Peak: 2.0342
      PeakTime: 1.4226e-07

```

But for $gm1=gm2=1 \times 10^{-3}$, the output of the program meets the requirements for the amplifier:

For gm1=1000 (uA/v)

Vx=0.05 (v); Gain_1st_Stage=44.67; Opt_Vout1=1.00 (v); Vicm=0.48 (v); I1=58.82 (uA)

Wn1=31.65 (uM); Ln1=0.50 (uM); Wp1=2.33 (uM); Lp1=1.00 (uM);

For gm2=1000 (uA/v)

Gain_2nd_Stage=49.06; Opt_Vout=0.95 (v); I2=58.82 (uA)

Wn2=0.91 (uM); Ln2=0.50 (uM); Wp2=28.81 (uM); Lp2=1.00 (uM);

Gain margin= 8.95 (dB) at 421.77 (MHz); Phase Margin= 38.78° at 121.9 (MHz)

RiseTime: 6.7358e-08
TransientTime: 1.6323e-07
SettlingTime: 1.6323e-07
SettlingMin: -1.9977
SettlingMax: -1.8030
Overshoot: 0
Undershoot: 0
Peak: 1.9977
PeakTime: 2.2444e-07

```

For gm1=2000 (uA/v)

Vx=0.05 (v); Gain_1st_Stage=44.67; Opt_Vout1=1.00 (v); Vicm=0.48 (v); I1=117.65 (uA)

Wn1=63.31 (uM); Ln1=0.50 (uM); Wp1=4.65 (uM); Lp1=1.00 (uM);

For gm2=2000 (uA/v)

Gain_2nd_Stage=19.09; opt_n2_VGS=0.90 (v); I2=531.28 (uA)

Wn2=17.19 (uM); Ln2=0.50 (uM); Wp2=20.78 (uM); Lp2=1.00 (uM);

Current Mirror Design
I_ref=1.81 (uA); Ln_ref=10.00 (uM); Wn_ref=0.50 (uM)

Ln_B1=0.50 (uA); Wn_B1=21.92 (uM)

Gain margin= 8.23 (dB) at 602.66 (MHz); Phase Margin= 26.94° at 196.4 (MHz)

    RiseTime: 6.6742e-08
TransientTime: 1.6130e-07
SettlingTime: 1.6130e-07
SettlingMin: -1.9963
SettlingMax: -1.8045
Overshoot: 0
Undershoot: 0
    Peak: 1.9963
    PeakTime: 2.2239e-07

```

For the design of the bottom part of the first stage of the amplifier, according to the following formulas, if the transistors are in the saturation region, the current ratio of each branch to I_{ref} is equal to the ratio of the width to the length of the transistor for each branch.

To reduce the value of I_{ref} (for lower overall power consumption of the circuit), the width of the transistor under the current source was set to 500 nm, which is the minimum allowable value. For the transistor under the first stage, the width was set to 10 micrometers, which ensures that the required current for both the first and second stages is met. Since the transistor under the first stage is not in saturation, the exact 10-fold ratio is not achieved.

Current Mirror

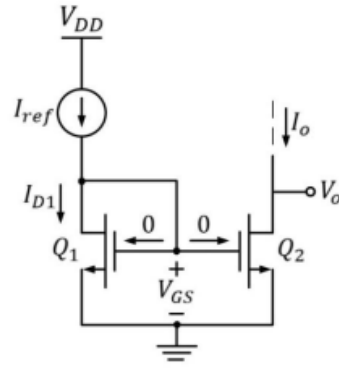
Basic MOSFET Current Mirror

Transistor Q_2 has the same V_{GS} as Q_1 . If Q_2 is set to operate in the **saturation** mode, its drain current will be

$$I_o = I_{D2} = \frac{1}{2} k'_n \left(\frac{W_2}{L_2} \right) (V_{GS} - V_{Tn})^2$$

The output current I_o is related to the reference current I_{ref} as

$$\frac{I_o}{I_{ref}} = \frac{I_{D2}}{I_{D1}} = \frac{W_2/L_2}{W_1/L_1} = m$$



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In testing the obtained values in Cadence Virtuoso using DC analysis, it was found that the voltage V_x became much smaller than 50 mV. However, by increasing V_{cm} from 480 mV to 600mV, V_x reached 50 mV.

Ultimately, the final parameters to be determined are as follows:

- W_{p1} : Width of the PMOS transistor in the first stage
- W_{n1} : Width of the NMOS transistor in the first stage
- W_{p2} : Width of the PMOS transistor in the second stage
- W_{n2} : Width of the NMOS transistor in the second stage
- I_{ref} : Current of the internal current source

Using Parametric Analysis around the initial values of W_{n1} , W_{n2} , W_{p1} , W_{p2} , and I_{ref} , the frequency response and DC bias of the amplifier are optimized so that the Phase Margin Frequency exceeds 100 MHz, while ensuring that the transistors are in the saturation region, the values of the mentioned parameters were obtained as shown in the figure below:

ADE L (2) - Project_lib tb_AC schematic

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Design Variables

	Name	Value
1	vddh	1.8
2	vocm	1.15
3	vddl	1.1
4	Vcm	600m
5	vcm	600m
6	Iref	60u
7	Wn1	25u
8	Wp2	18u
9	Wn2	4u

> Results in /home/disk/simulation/SIM_IC61/tb

Analyses

Type	Enable	Arguments
1 stb	☒	10 1G /IPRB0 Automatic Start-Stop

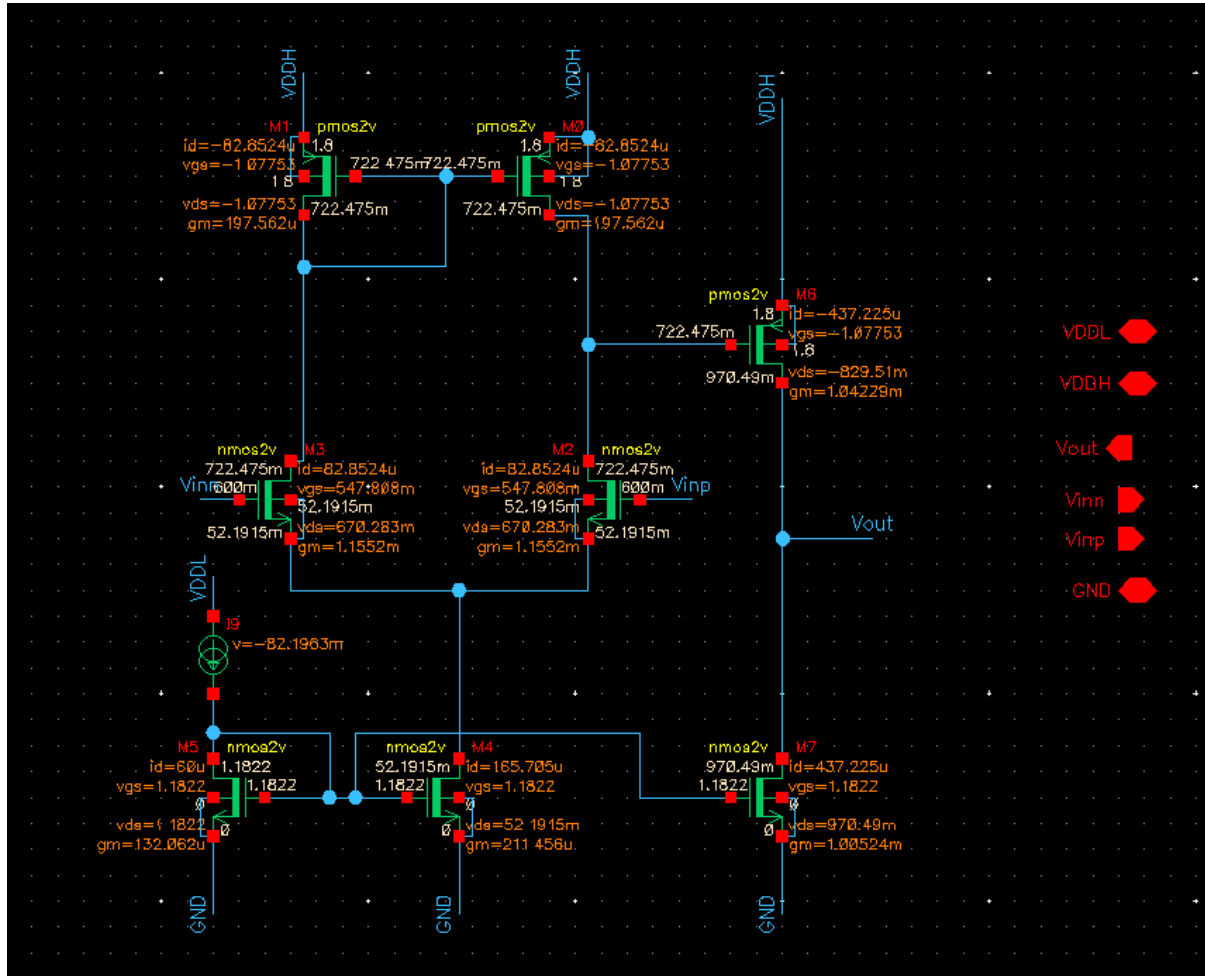
Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 Loop_Gain_dB20	wave	☒	☐	
2 Loop_Gain_Phase	wave	☒	☐	
3 Phase_Margin	56.555	☒	☐	
4 Phase Margin Frequency	106.776M	☒	☐	

Plot after simulation: Auto Plotting mode: Replace

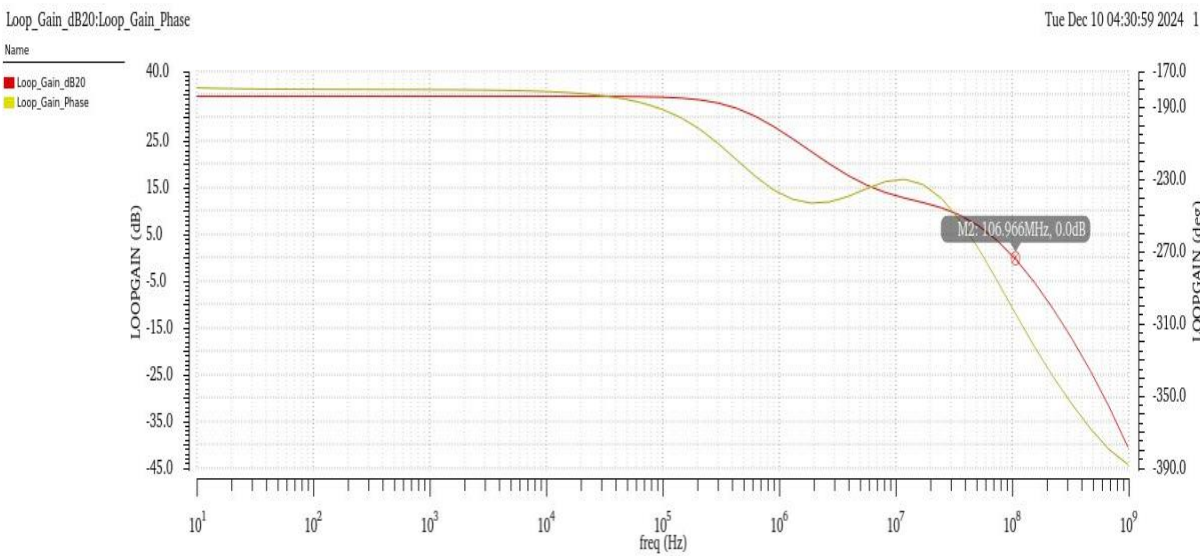
3(5) Save State ... Status: Ready T=27 C Simulator: spectre State: state1

DC Analysis:

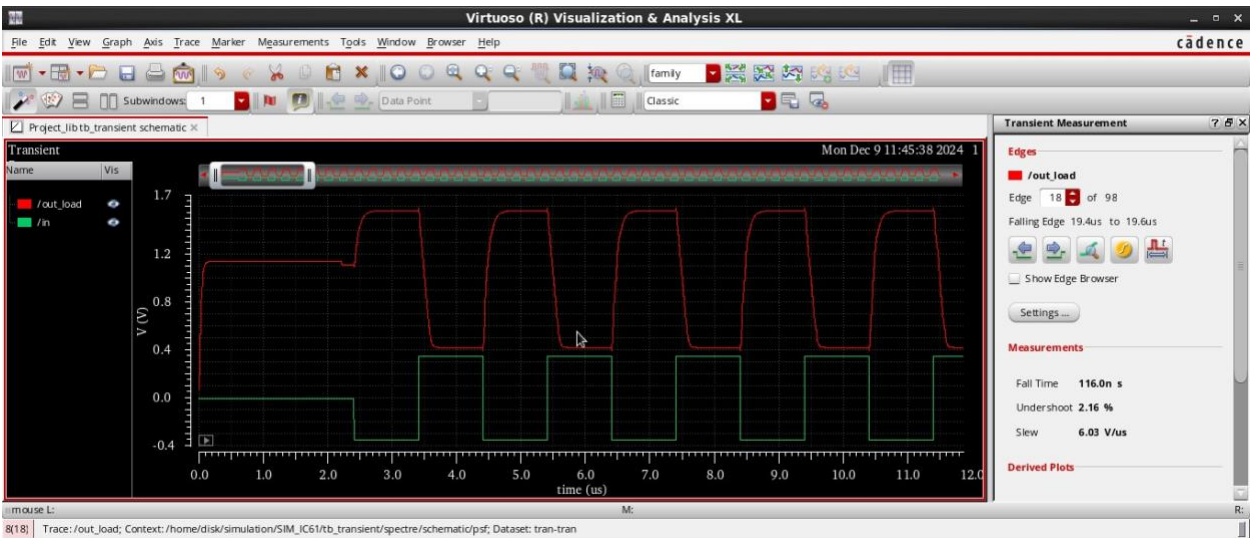
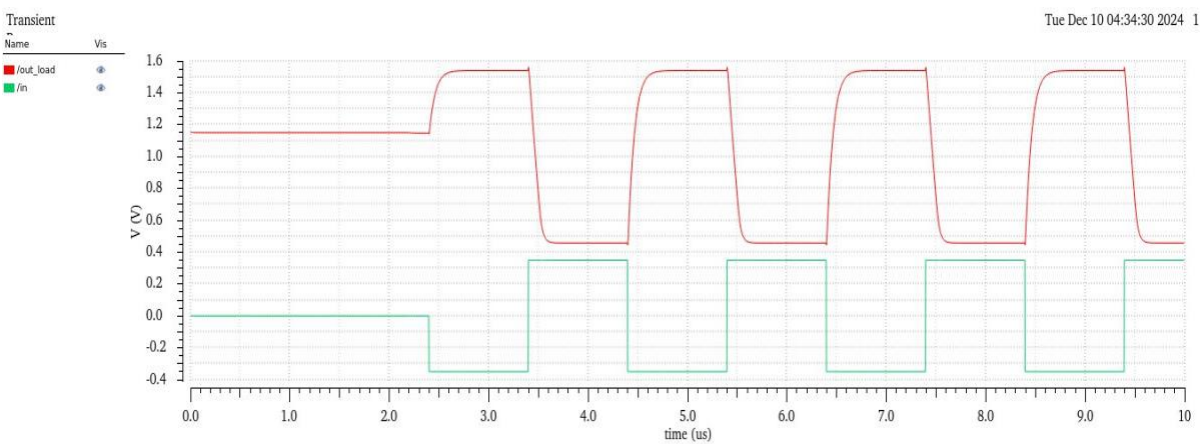


All transistors in the first and second stages are in the saturation region, and the power consumption has been obtained to be less than 1.5 mW.

Phase Margin Analysis:



Transient Analysis:



The gain of 2 is maintained even when the input has a 700 mV amplitude, and the settling time (T_{settle}) is less than 100 ns.

Virtuoso® ADE Explorer Editing: project lib tb_AC maestro

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Setup

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Parameters

Reliability Analyses

Monte Carlo Sampling

Checks/Asserts

Name	Value	Name	Type	Details	Value	Plot	Save	Spec
CM_Gain	expr	vfreq/ac "vcm_out"						
DM_Gain	expr	vfreq/ac "vdm_out"						
DM_Gain_at_DC	expr	magvalue(DM_Gain 10)			312.7			
CMRR	expr	dB20(DM_Gain / CM_Gain)						
CMRR_at_DC	expr	value(CMRR 10)			20.72			
VDDH_Gain	expr	vfreq/ac "vps_out_vddh"						
PSRR_VDDH	expr	dB20(DM_Gain / VDDH_Gain)						
PSRR_VDDH_at_DC	expr	value(PSRR_VDDH 10)			-6.18K			
PSRR_VDDH_at_DC	expr	value(PSRR_VDDH 10)			49.07			
Loop_Gain_Phase	expr	phaseDegUnwrappedGetData("loopGain" "result" "tb")						
Loop_Gain_at_20	expr	[dB (magGetData("loopGain" "result" "tb") 10)]			eval err			
Phase Margin	expr	getData("phaseMargin" "result" "tb_margin")			49.25			
Phase Margin Frequency	expr	getData("phaseMarginFreq" "result" "tb_margin")			101.4M			

Results: ExplorerRun.0 project lib tb_AC schematic Simulator: spectre app Batch

Virtuoso® ADE Explorer Editing: project lib tb_transient maestro

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Setup

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Outputs Setup

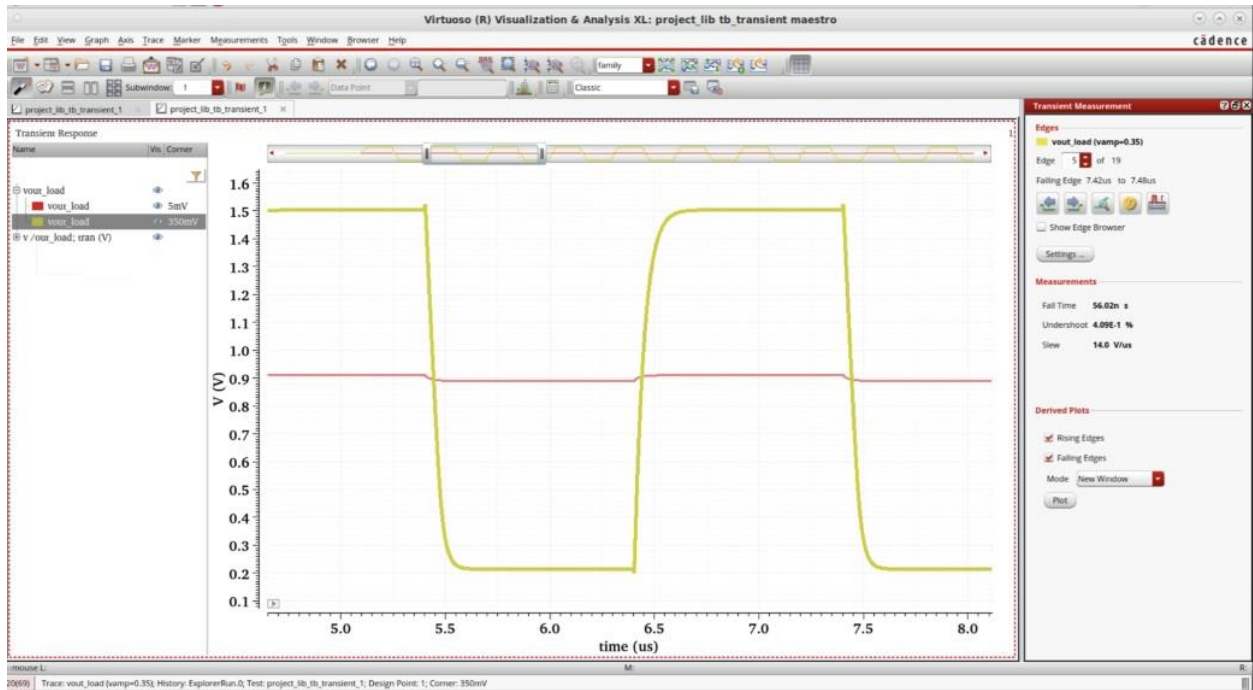
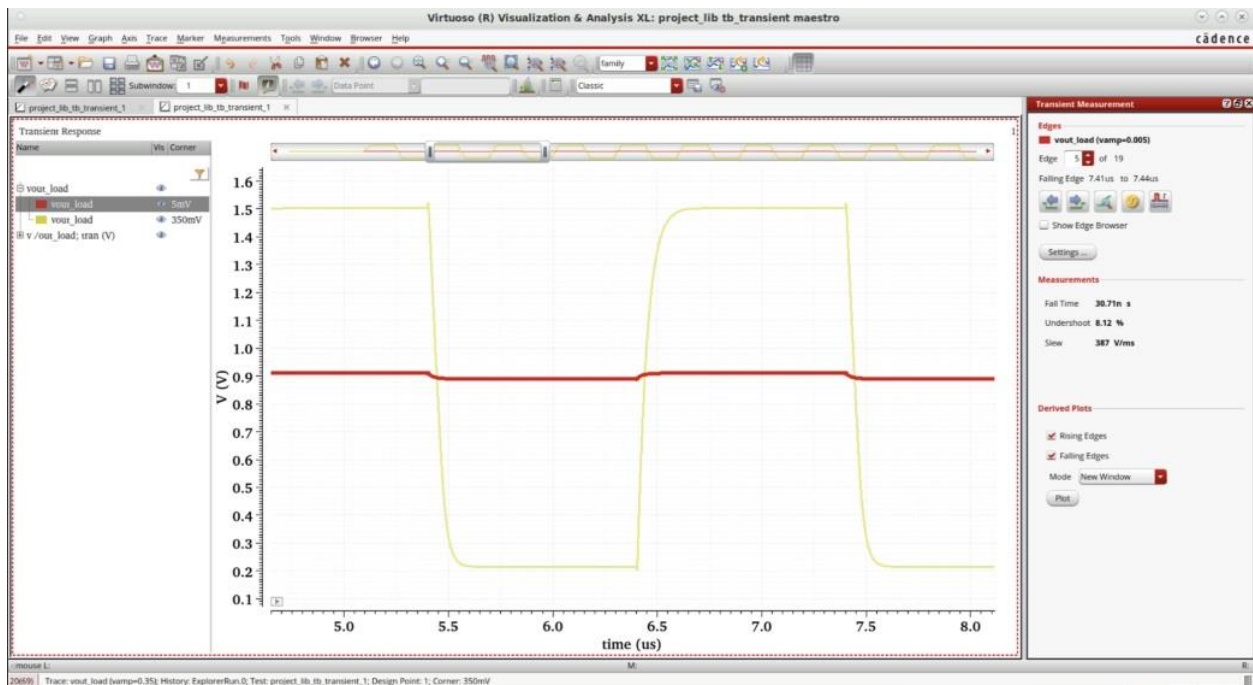
Results

Parameter	5mV	350mV
vamp	5m	350m

8/10 rows

Test	Output	Spec	Weight	Pass/Fail	Min	Max	5mV	350mV
project lib tb tr...	vout_load				1.753m	2.023m	2.023m	1.753m
project lib tb tr...	avg_prow_108							
project lib tb tr...	err_high				219.5n	219.5n	219.5n	eval err
project lib tb tr...	rise_time				207.7n	207.7n	207.7n	eval err
project lib tb tr...	err_low				219.5n	219.5n	219.5n	eval err
project lib tb tr...	fall_time				2.252	2.252	2.252	eval err
project lib tb tr...	settle_time							
project lib tb tr...	Fold							

Results: ExplorerRun.0 project lib tb_transient schematic Simulator: spectre



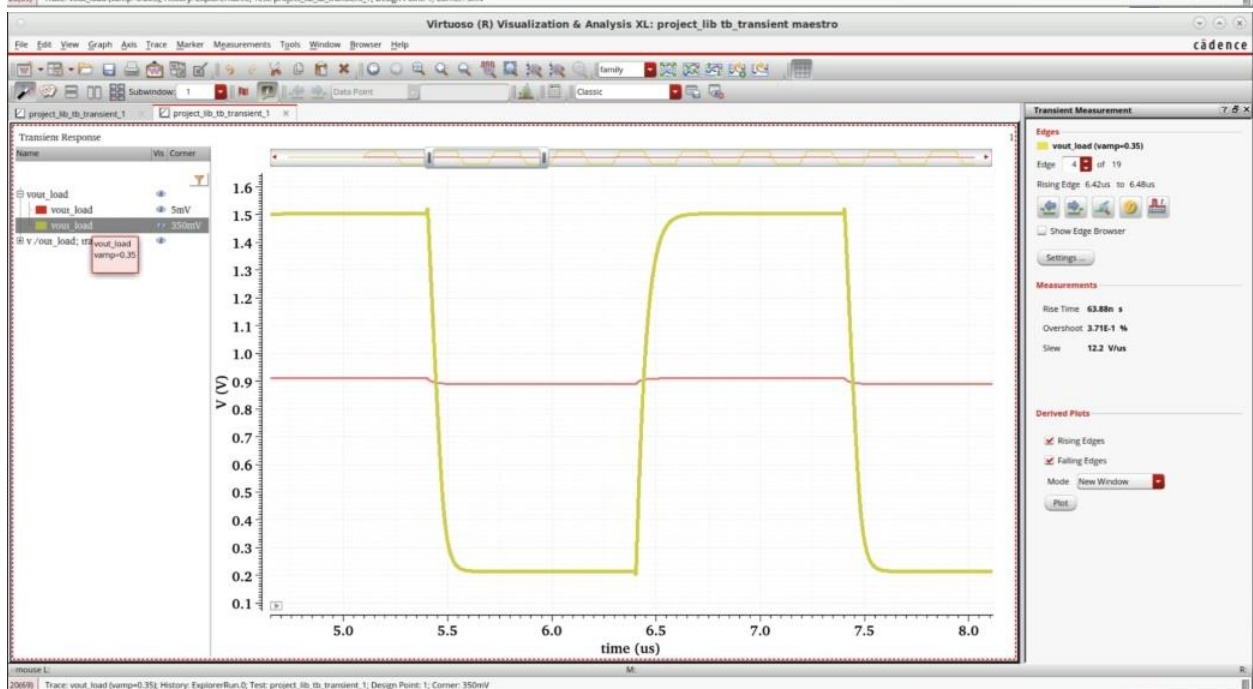
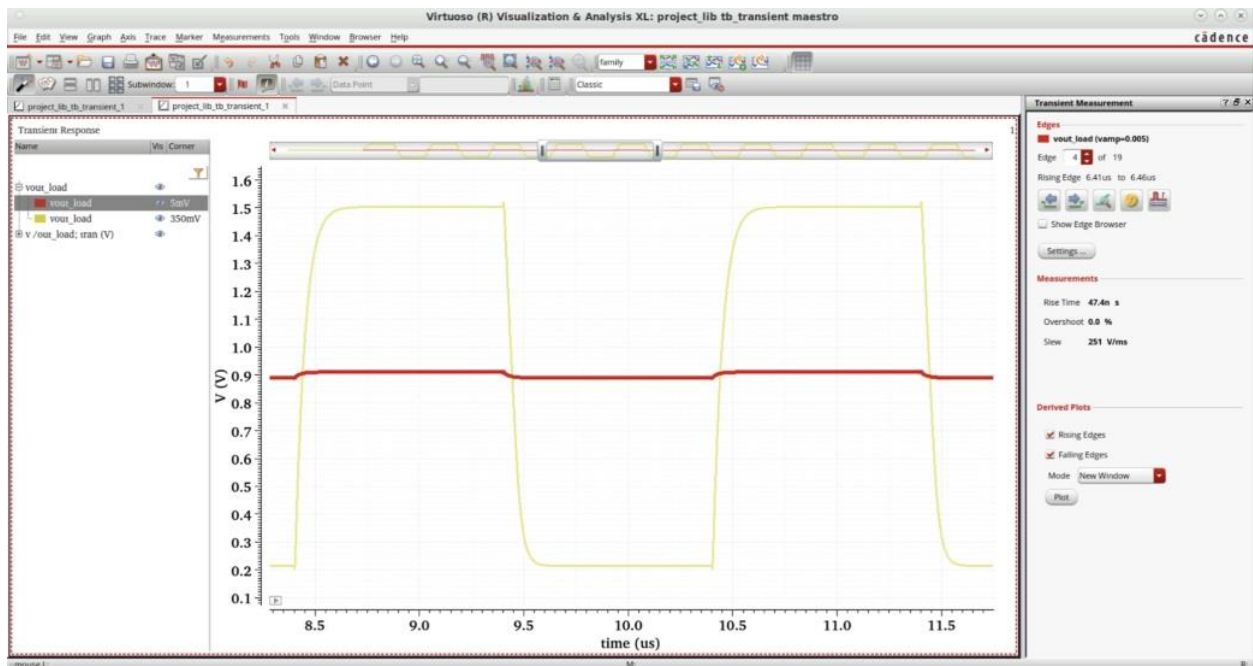


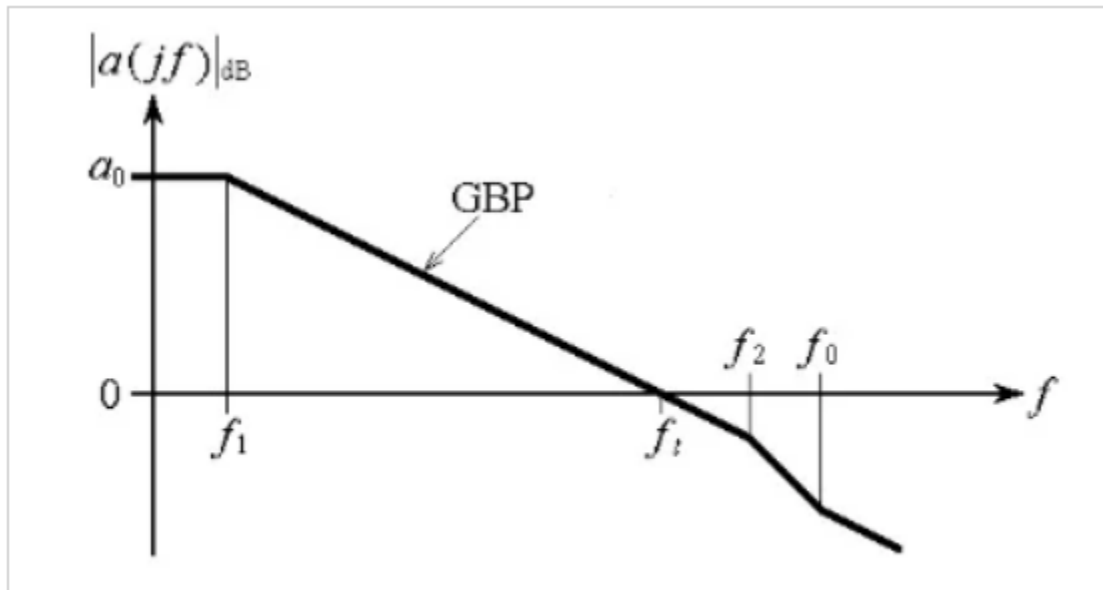
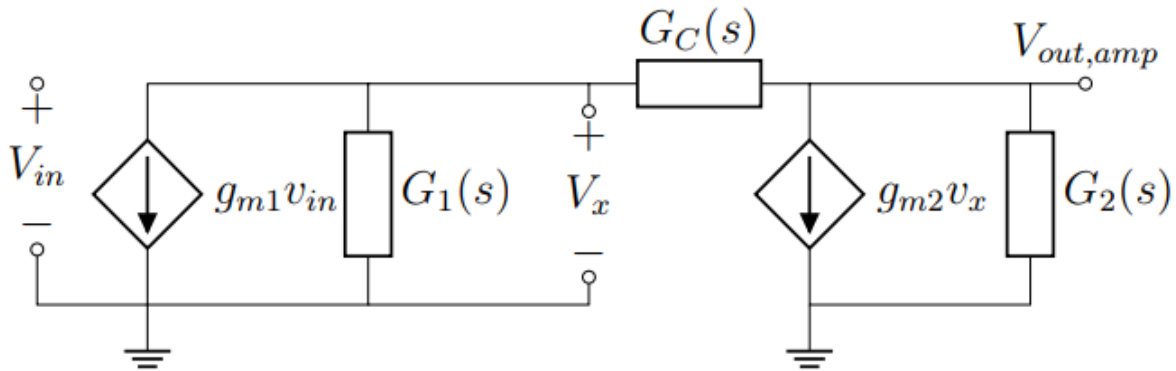
Table:

Closed loop DC gain	2
Settling time	1.9954×10^{-7}
Power consumption	2.023 mW
CMRR at DC	20.72
PSRR VDDH at DC	49.07
PSRR VDDL at DC	-6.18

Conclusion:

The project aims to design an amplifier that meets specific performance criteria, including a maximum settling time of less than 180 ns and power consumption below 1.5 mW. The amplifier is optimized to maximize the figure of merit (FOM), which is the inverse of power consumption multiplied by settling time. The design employs a cascode topology, chosen for its simplicity, higher voltage headroom, and ability to maintain transistor saturation. MATLAB code was used for initial design and symbolic analysis, with adjustments made for parameters such as transistor widths and currents to meet the required performance. In simulations using Cadence Virtuoso, all transistors were confirmed to be in the saturation region, and the final design maintained a phase margin exceeding 100 MHz. The amplifier's gain of 2 and settling time of under 100 ns were achieved, even with a 700 mV input amplitude.

PoleZero_Formula.docx:



Gain magnitude profile of a Miller compensated op-amp.

- The *dc gain* a_0
- The *pole-frequency pair* f_1 and f_2 , with f_1 being the *dominant pole*
- The *transition frequency* f_t at which gain crosses the 0 dB axis
- The *zero frequency* f_0 associated with the RHPZ

$$V_{out}|_{f_0} = 0, \text{ So } V_x * G_c(s) = g_{m2} * V_x \gg C_m * \omega_0 = g_{m2}$$

$$\omega_0 = \frac{g_{m2}}{C_m}$$

Base on Behavioral Model

$$\omega_2 = \frac{g_{m2}}{(C_L || C_s || C_f) + C_o}$$

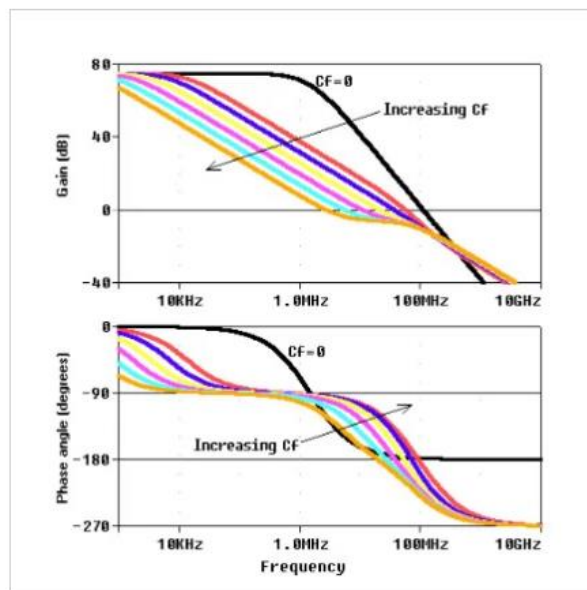
$$\omega_1 = \frac{1}{R_1[C_1 + C_f(1 + G_{m2}R_2 + R_2/R_1)] + R_2C_2}$$

Equation 3

Retaining only the dominant portion, we approximate as

$$\omega_1 \cong \frac{1}{R_1 R_2 g_{m2} C_m}$$

It is apparent that the zero frequencies of the magnitude curves are just too close to the corresponding transition frequencies to allow the designer much flexibility in achieving acceptable phase margins.



Matlab Code:


```

% Exercise 5a: Common-source with active PMOS load

clc

clear

nch = importdata ('nch_1v.mat');
pch = importdata ('pch_1v.mat');

%%

fu = 1e7;

ft = fu *10;

CL = 1e-12;

VDD = 1.8;

Vx=0.1; % voltage of Current Mirror

Desired_n_gm=600e-6; %desired gm for achieve phase margin and fu

L_range = nch.L;

gm_ID_range = 2:1:30;

Vout_range = Vx+(0.05:0.05:1.1);

% Sweep over Vout

for i = 1: length ( Vout_range )

    % Find the best PMOS L

    % My code to look up pmos gds /ID across l given VGS ,VDS

    p_gds_id_vs_l = look_up(pch,'GDS_ID','L', L_range,'VDS',VDD-Vout_range(i),'VGS',VDD-
Vout_range(i));

    % My code to look up pmos ft across l given VGS ,VDS

    p_ft_vs_l =look_up (pch , 'GM_CGG', 'L', L_range,'VDS',VDD-Vout_range(i),'VGS',VDD-
Vout_range(i))/2/pi;

    M = p_ft_vs_l > ft;

    if ( any(M))

        % We know gds_id will decrease as L increases , so find the highest L

        best_idx = max( find (M ==1) );

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p_gds_id = p_gds_id_vs_l ( best_idx );
best_p_l (i) = L_range ( best_idx );
% Knowing PMOS GDS and L, we can optimize the NMOS L and gm_ID
for j = 1: length ( L_range )
    % Find the gm_ID that maximizes overall gain while meeting ft constraint
    % My code to find the NMOS gds_id across the gm_ID_range
    n_gds_id =
look_up(nch,'GDS_ID','GM_ID',gm_ID_range,'L',L_range(j),'VDS',Vout_range(i)-Vx);
    % My code to find the NMOS ft across the gm_ID_range
    n_ft =
look_up(nch,'GM_CGG','GM_ID',gm_ID_range,'L',L_range(j),'VDS',Vout_range(i)-Vx)/2/pi;
    % My code to find Av from gm_ID_range , n_gds_id and p_gds_id
    av = gm_ID_range./(n_gds_id'+p_gds_id);
    K = n_ft > ft;
    if (any (K))
        max_av = max (av( find (K ==1) ));
        best_idx = find (av == max_av );
        best_n_gm_ID (i, j) = gm_ID_range ( best_idx );
        best_Av (i, j) = av( best_idx );
    else
        best_n_gm_ID (i,j) = NaN ;
        best_Av (i,j) = NaN ;
    end
end
else
    best_n_gm_ID (i, :) = NaN ;
    best_Av (i, :) = NaN ;
    best_p_l (i) = NaN;

```

```

end

end

% Find the best Av
max_gain = max(max( best_Av ));
[ Vout_idx , n_L_idx ] = find ( best_Av == max_gain );
opt_p_L = best_p_l ( Vout_idx );
opt_n_gm_ID = best_n_gm_ID ( Vout_idx , n_L_idx );
% My code to find the optimum nmos L
opt_n_L = L_range(n_L_idx);
% find the optimal Vout
opt_Vout = Vout_range(Vout_idx);
%*****

% Find the real widths of the devices
if opt_Vout>Vx
    % My code to find the nmos CDD /W at the optimum gm_ID, L, VDS
    n_cdd_w = look_up (nch ,'CDD_W', 'GM_ID', opt_n_gm_ID , 'L', opt_n_L, 'VDS', opt_Vout-Vx);

    % My code to find the nmos JD at the optimum gm_ID , L, VDS
    n_JD = look_up (nch , 'ID_W', 'GM_ID', opt_n_gm_ID , 'L', opt_n_L, 'VDS', opt_Vout-Vx);

    % My code to find the pmos CDD /W at the optimum VGS , VDS , L
    p_cdd_w = look_up (pch ,'CDD_W', 'L', opt_p_L, 'VDS', VDD-opt_Vout, 'VGS', VDD-opt_Vout);

    % My code to find the pmos JD at the optimum VGS , VDS , L
    p_JD = look_up (pch , 'ID_W', 'L', opt_p_L, 'VDS', VDD-opt_Vout, 'VGS', VDD-opt_Vout);

    I=Desired_n_gm/opt_n_gm_ID;
    Wn=I/n_JD;

```

```

Wp=I/p_JD;

n_VGS=look_upVGS(nch,'GM_ID',opt_n_gm_ID,'L',opt_n_L,'VDS',opt_Vout-Vx);
Vicm=n_VGS+Vx;

fprintf("Vx=%.2f(v); max_Gain=%.2f; Opt_Vout=%.2f(v); Vicm=%.2f(v);
I0=%.2f(uA)\r\n",Vx,max_gain,opt_Vout,Vicm,2*I*1e6);

fprintf("Wn=%.2f(uM); Ln=%.2f(uM); Wp=%.2f(uM); Lp=%.2f(uM);\r\n",...
        Wn,opt_n_L,Wp,opt_p_L);

disp("");

end

%*****

```